

CLASS TEST 3

MEMO

01
$$\begin{aligned}
 v(t) &= 10 \sin(100t + 60^\circ) \\
 &= 10 \cos(100t + 60^\circ - 90^\circ) \\
 &= 10 \cos(100t - 30^\circ) \text{ V} \\
 \therefore V &= 10 \angle -30^\circ \text{ V}
 \end{aligned}$$

02 1 (v_1 lags v_2)

03
$$\begin{aligned}
 Z_{ab} &= \frac{(5 + \frac{1}{j0.5})(j10 + \frac{1}{j0.1})}{5 + \frac{1}{j0.5} + j10 + \frac{1}{j0.1}} \\
 &= 0 \Omega
 \end{aligned}$$

04-05
$$\begin{aligned}
 V_o &= \frac{10}{10 + j50(0.1)} V_g \\
 &= \frac{10}{10 + j5} 20 = 16 - j8 = 17.89 \angle -26.56^\circ \text{ V} \\
 \therefore v_o(t) &= \operatorname{Re} \{ V_o e^{j50t} \} \\
 &= 17.85 \cos(50t - 26.56^\circ) \text{ V}
 \end{aligned}$$

04 $B = 17.85 \text{ V}$

05 $C = -26.56^\circ$

06 KCL at 1:

$$\frac{V_1 - (-10)}{20} + \frac{V_1}{j4} - 2V_1 = 0$$

$$V_1 \left(\frac{1}{20} + \frac{1}{j4} - 2 \right) = -\frac{10}{20}$$

$$\therefore V_1 = 0.2523 - j0.03234 \text{ V}$$

07 $V_{Th} = \frac{(-j10) \parallel 20}{j5 + (-j10) \parallel 20} 100 \angle 45^\circ$ (voltage division)

$$= 178.9 \angle 18.43^\circ \text{ V}$$

08 $Z_{Th} = \left(\frac{1}{j5} + \frac{1}{-j10} + \frac{1}{20} \right)^{-1}$

$$= 4 + j8 \Omega$$

09 $\omega = 10 \text{ rad/s} :$

$$\therefore I' = \frac{8 \angle 30^\circ}{20 + j10(0.3)} = 0.3956 \angle +21.47^\circ \text{ A}$$

10 $\omega = 4 \text{ rad/s} :$

$$I'' = - \frac{3 \angle 0^\circ}{20 + j4(0.3)} = 0.1477 \angle 176.6^\circ \text{ A}$$